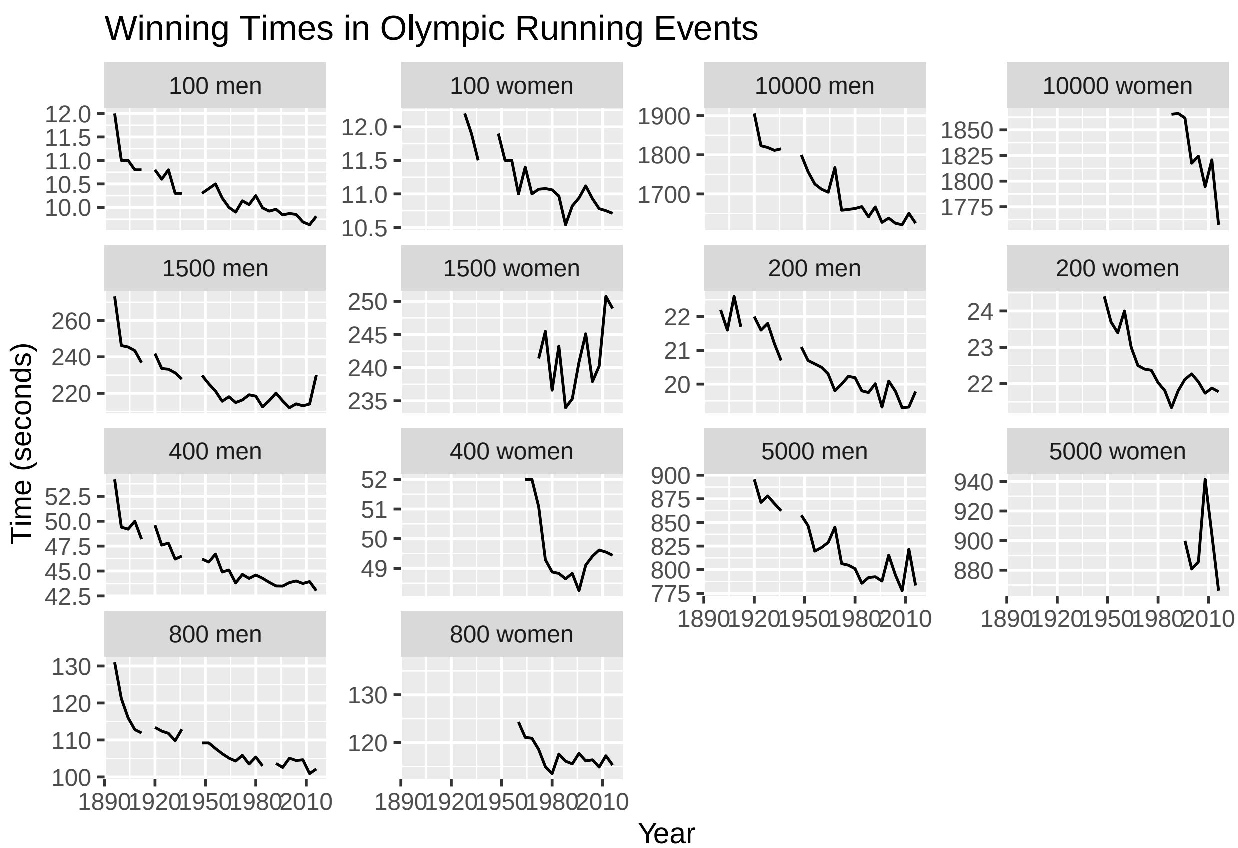
Olympic Running Times Analysis

Applied Forecasting - Spring 2025

Question 1: Olympic Running Times

1.1 Plot the winning time against the year for each event (Length + Sex)

```
# Load the dataset
data("olympic_running")

# Create Event variable
data <- olympic_running %>%
  mutate(Event = paste(Length, Sex))

# Plot winning times by event
ggplot(data, aes(x = Year, y = Time)) +
  geom_line() +
  facet_wrap(~Event, scales = "free_y") +
  labs(title = "Winning Times in Olympic Running Events",
       y = "Time (seconds)", x = "Year")
```



1.2 Fit a regression line to each event

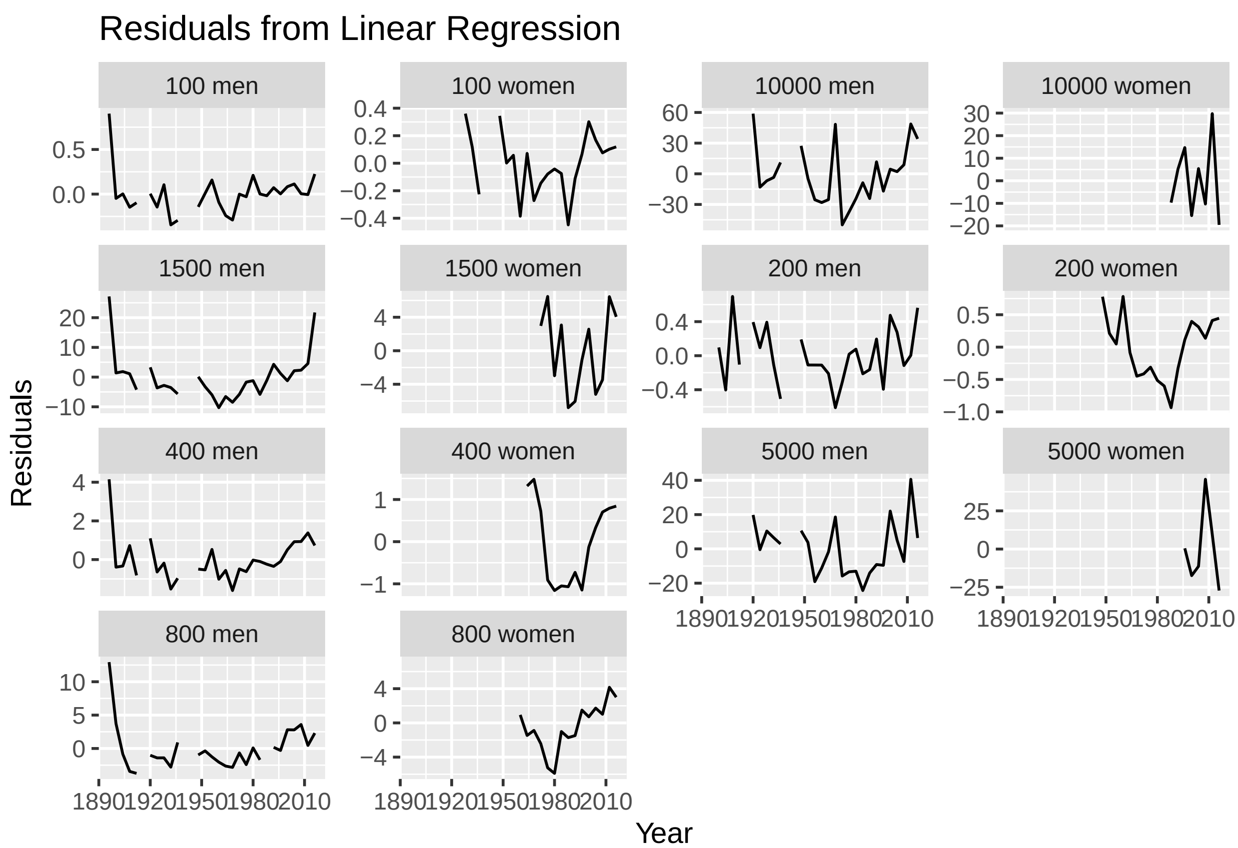
```r
# Fit linear regression model by Event
lm_models <- data %>%
  model(lm_model = TSLM(Time ~ Year))

# Extract slope coefficients
lm_models %>%
  tidy() %>%
  filter(term == "Year")
```

```
## # A tibble: 14 x 8
##    Length Sex   .model   term  estimate std.error statistic  p.value
##     <int> <chr> <chr>    <chr>    <dbl>     <dbl>     <dbl>    <dbl>
##  1    100 men   lm_model Year   -0.0126   0.00120    -10.5  7.24e-11
##  2    100 women lm_model Year   -0.0142   0.00187     -7.58 3.72e- 7
##  3    200 men   lm_model Year   -0.0249   0.00181    -13.7  3.80e-13
##  4    200 women lm_model Year   -0.0336   0.00568     -5.92 2.17e- 5
##  5    400 men   lm_model Year   -0.0646   0.00586    -11.0  2.75e-11
##  6    400 women lm_model Year   -0.0401   0.0170      -2.35 3.65e- 2
##  7    800 men   lm_model Year   -0.152    0.0177      -8.58 6.47e- 9
##  8    800 women lm_model Year   -0.198    0.0384      -5.16 1.45e- 4
##  9   1500 men   lm_model Year   -0.315    0.0418      -7.54 5.23e- 8
## 10   1500 women lm_model Year    0.147    0.106        1.38 1.96e- 1
```

```
## 11   5000 men   lm_model Year   -1.03     0.104      -9.88  1.50e- 9
## 12   5000 women lm_model Year   -0.303    1.73       -0.175 8.69e- 1
## 13  10000 men   lm_model Year   -2.67     0.192     -13.9   2.37e-12
## 14  10000 women lm_model Year   -3.50     0.701      -4.99  2.47e- 3
```

1.3 Plot the residuals

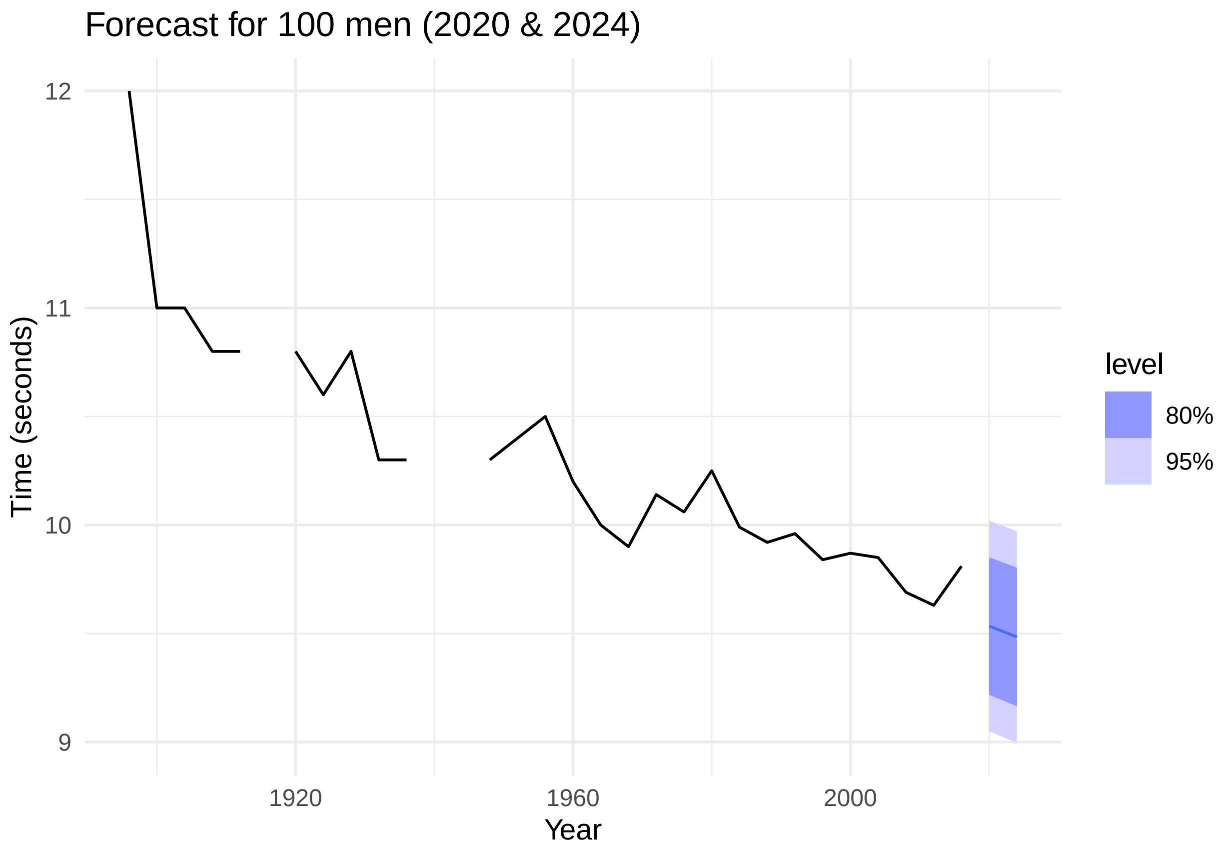
```r
library(fpp3)

# Reload the dataset from scratch to avoid accidental mutation
data("olympic_running")

# Step 1: Create an Event variable and convert to grouped tsibble
olympic_grouped <- olympic_running %>%
  mutate(Event = paste(Length, Sex)) %>%
  as_tsibble(key = Event, index = Year)

# Step 2: Fit a regression model for each event
lm_models <- olympic_grouped %>%
  model(lm_model = TSLM(Time ~ Year))

# Step 3: Augment + Residual Plot
lm_models %>%
  augment() %>%
  ggplot(aes(x = Year, y = .resid)) +
  geom_line() +
  facet_wrap(~Event, scales = "free_y") +
  labs(title = "Residuals from Linear Regression",
       y = "Residuals", x = "Year")
```

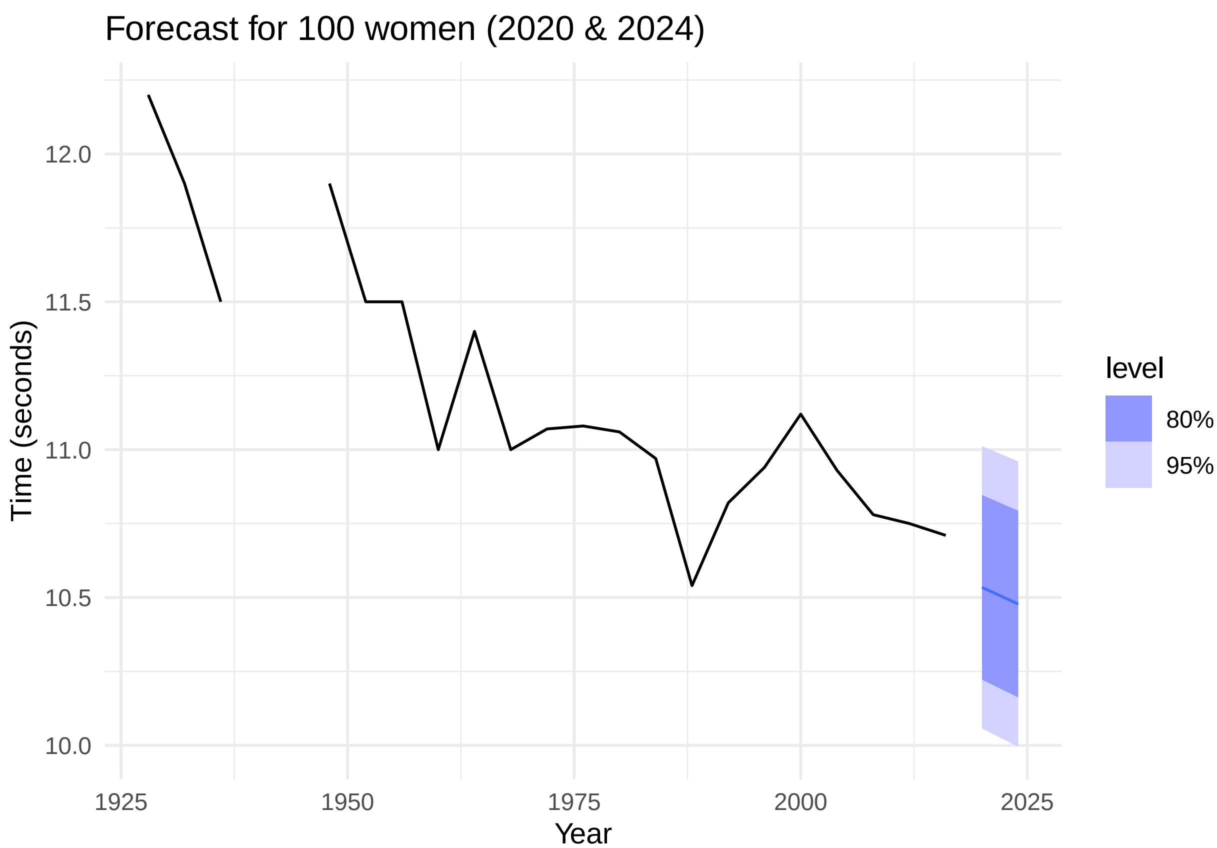


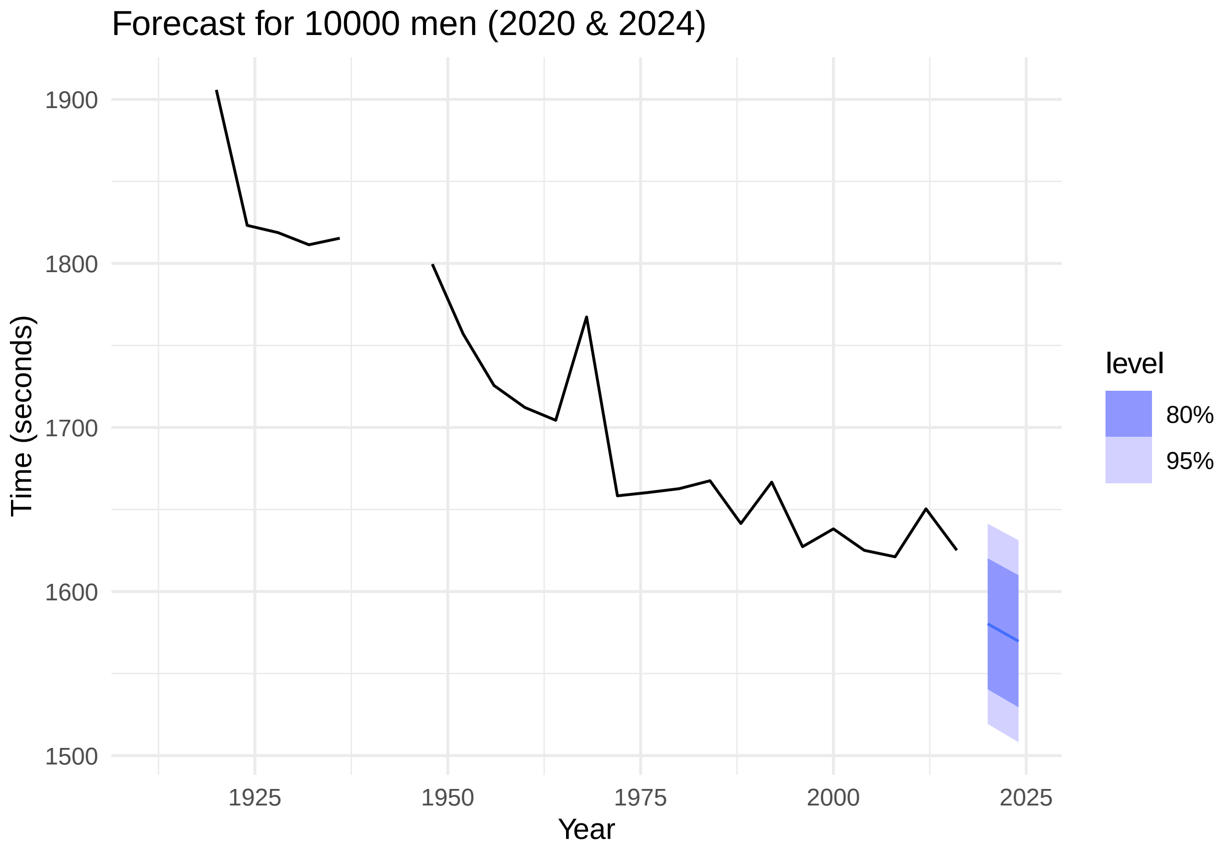
1.4 Forecast for 2020 and 2024

```
# Step 1: Get future years and combine with keys
future_olympics <- olympic_grouped %>%
  distinct(Event) %>%
  crossing(Year = c(2020, 2024)) %>%
  as_tsibble(index = Year, key = Event)

# Step 2: Forecast using the model
forecasts <- lm_models %>%
  forecast(new_data = future_olympics)

# Step 3: Plot forecasts
event_list <- unique(olympic_grouped$Event)

for (e in event_list) {
  print(
    forecasts %>%
      filter(Event == e) %>%
      autoplot(olympic_grouped %>% filter(Event == e)) +
      labs(title = paste("Forecast for", e, "(2020 & 2024)"),
           y = "Time (seconds)", x = "Year") +
      theme_minimal()
  )
```

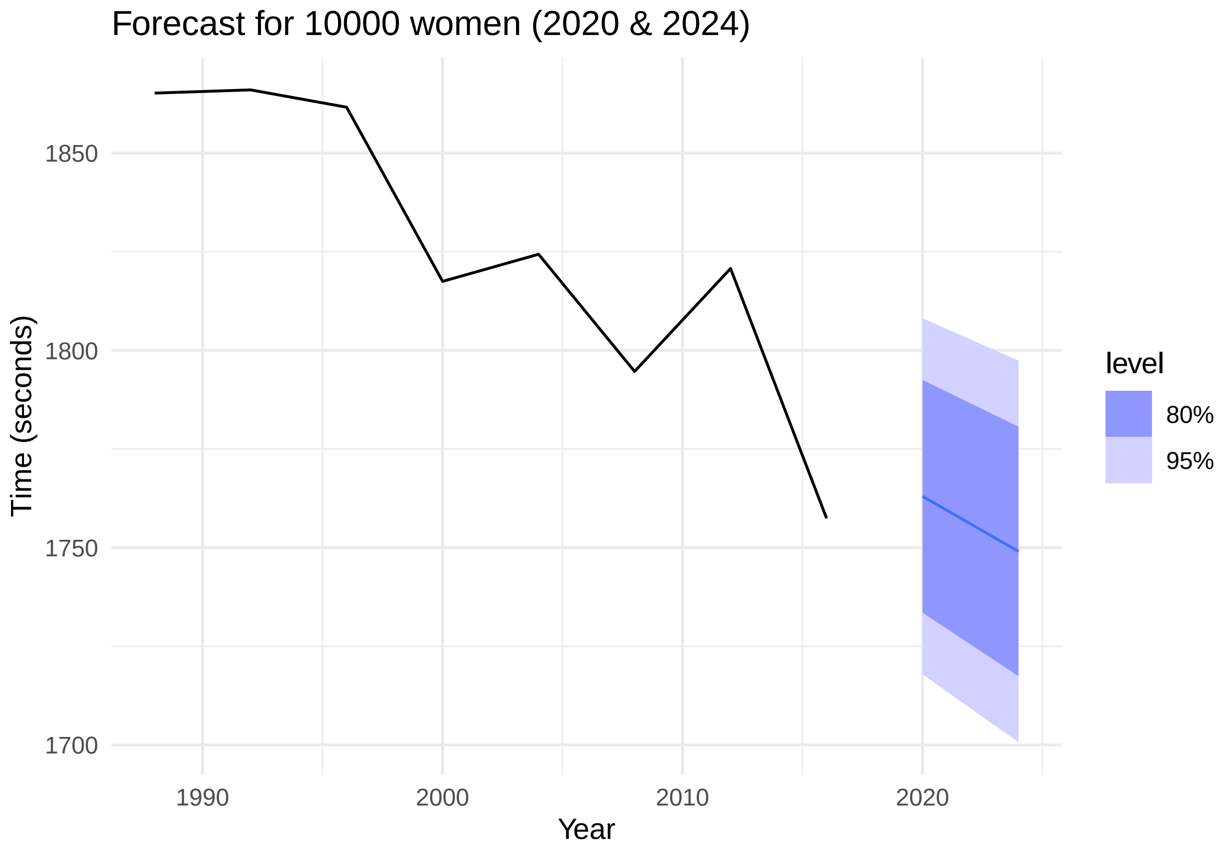
```
}
```


Forecast for 100 men (2020 & 2024)
12
11
10
9
Time (seconds)
1920
1960
2000
Year
level
80%
95%

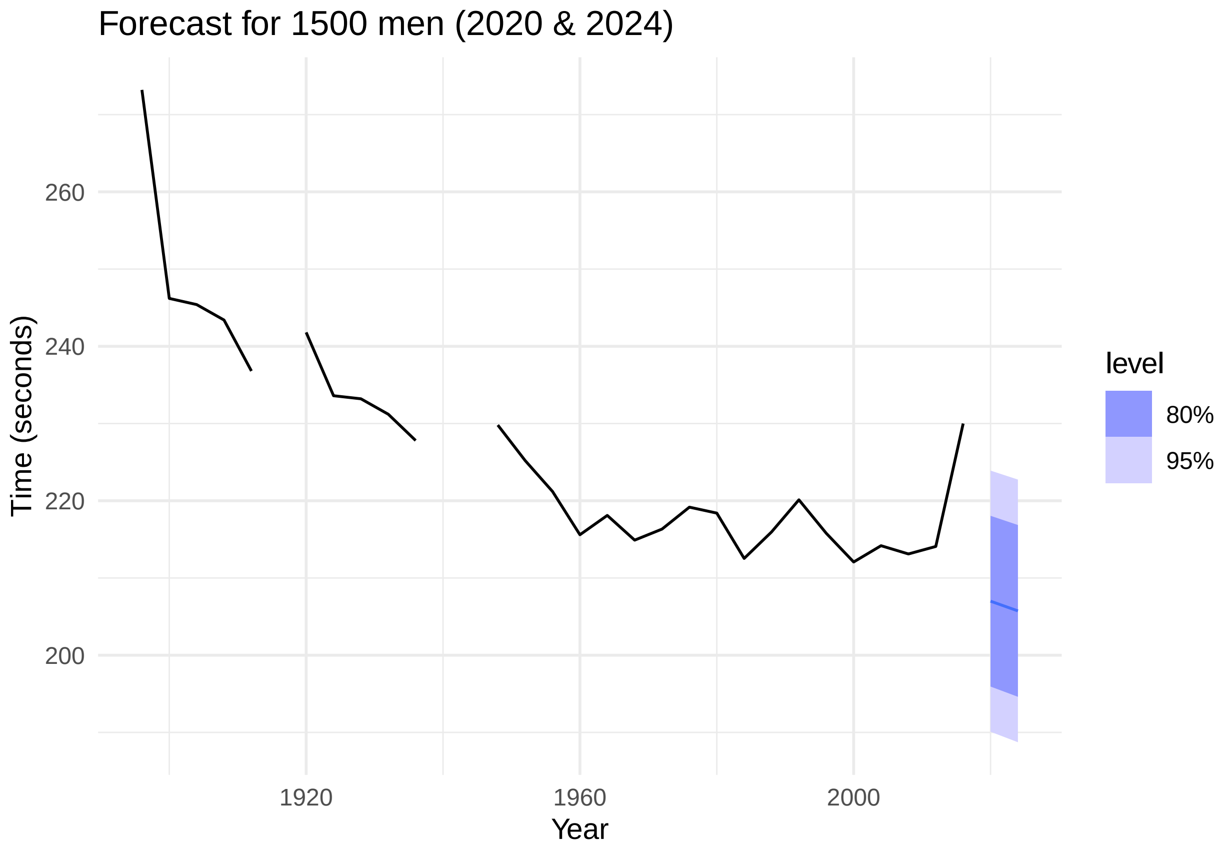

Forecast for 100 women (2020 & 2024)
Time (seconds)
12.0
11.5
11.0
10.5
10.0
1925
1950
1975
2000
2025
Year
level
80%
95%

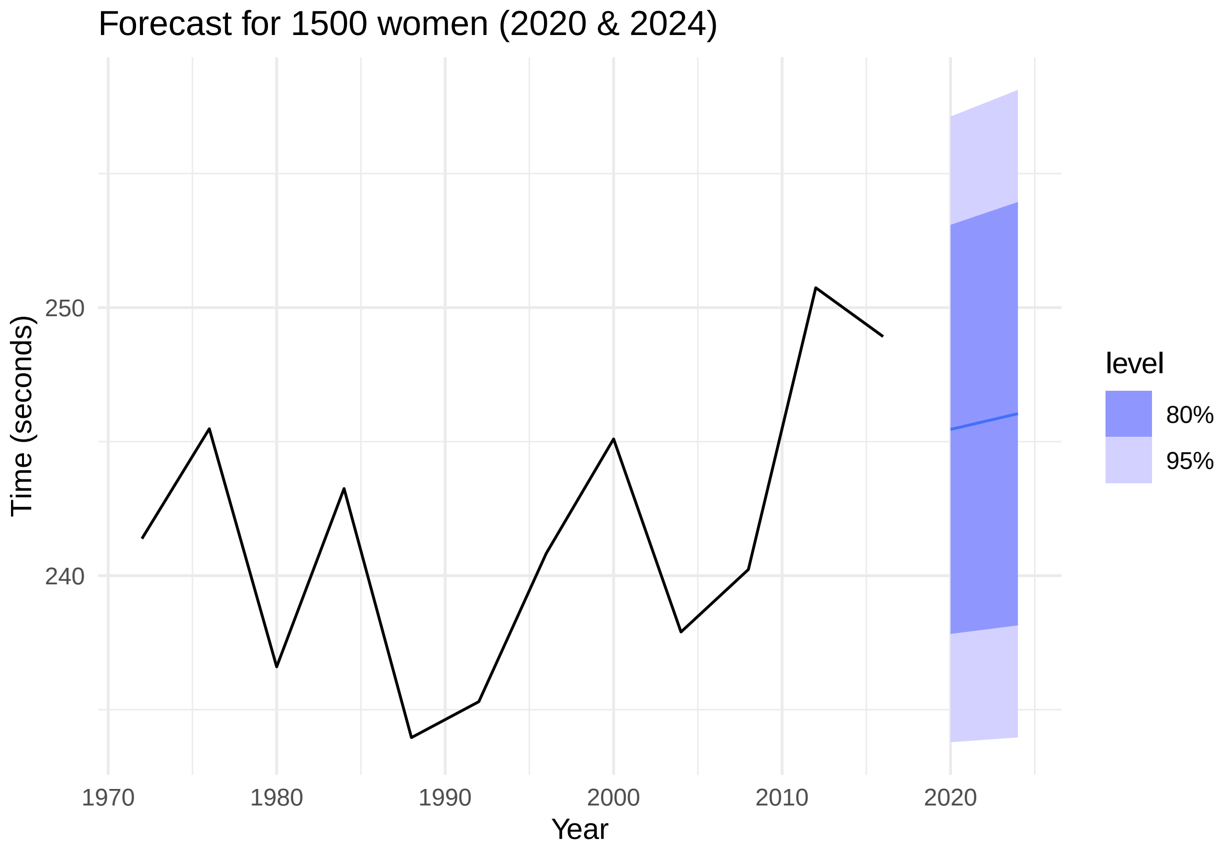


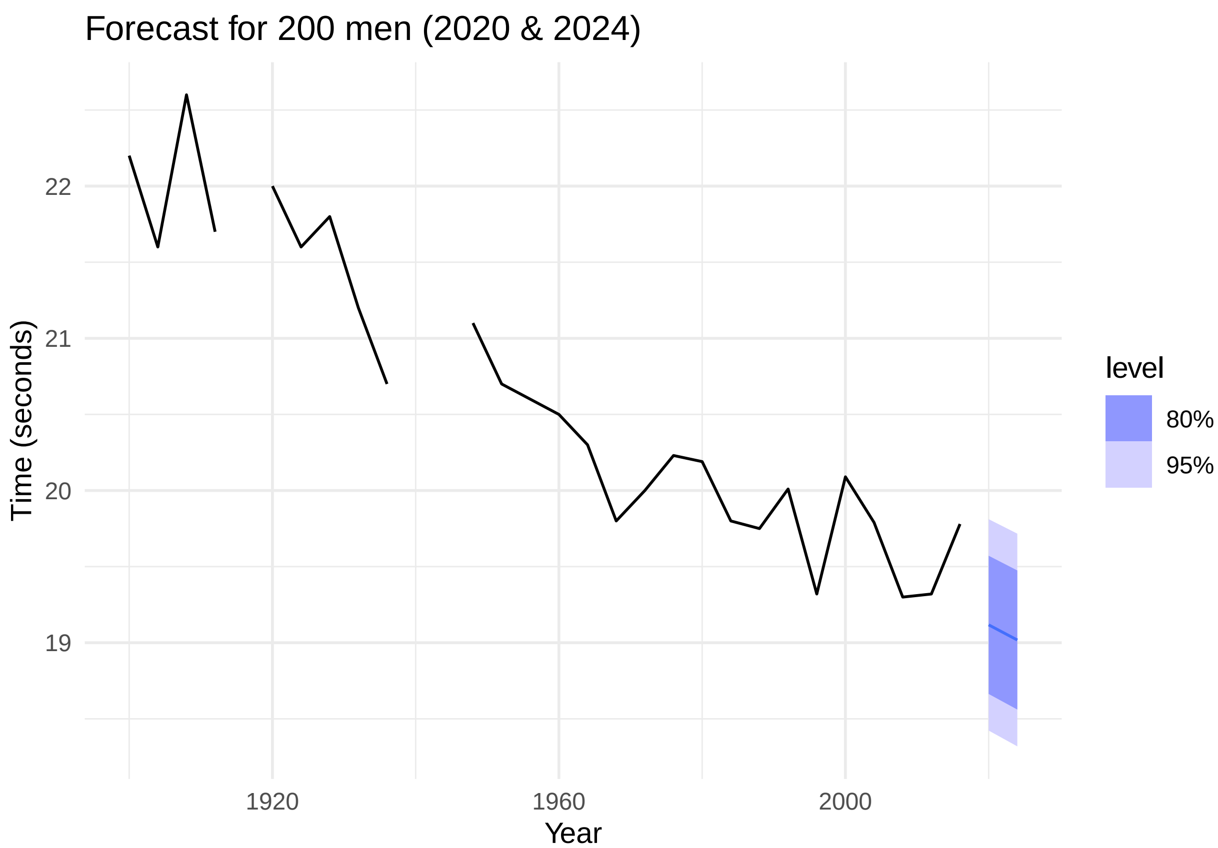
Forecast for 10000 men (2020 & 2024)
1900
1800
1700
1600
1500
Time (seconds)
1925
1950
1975
2000
2025
Year
level
80%
95%

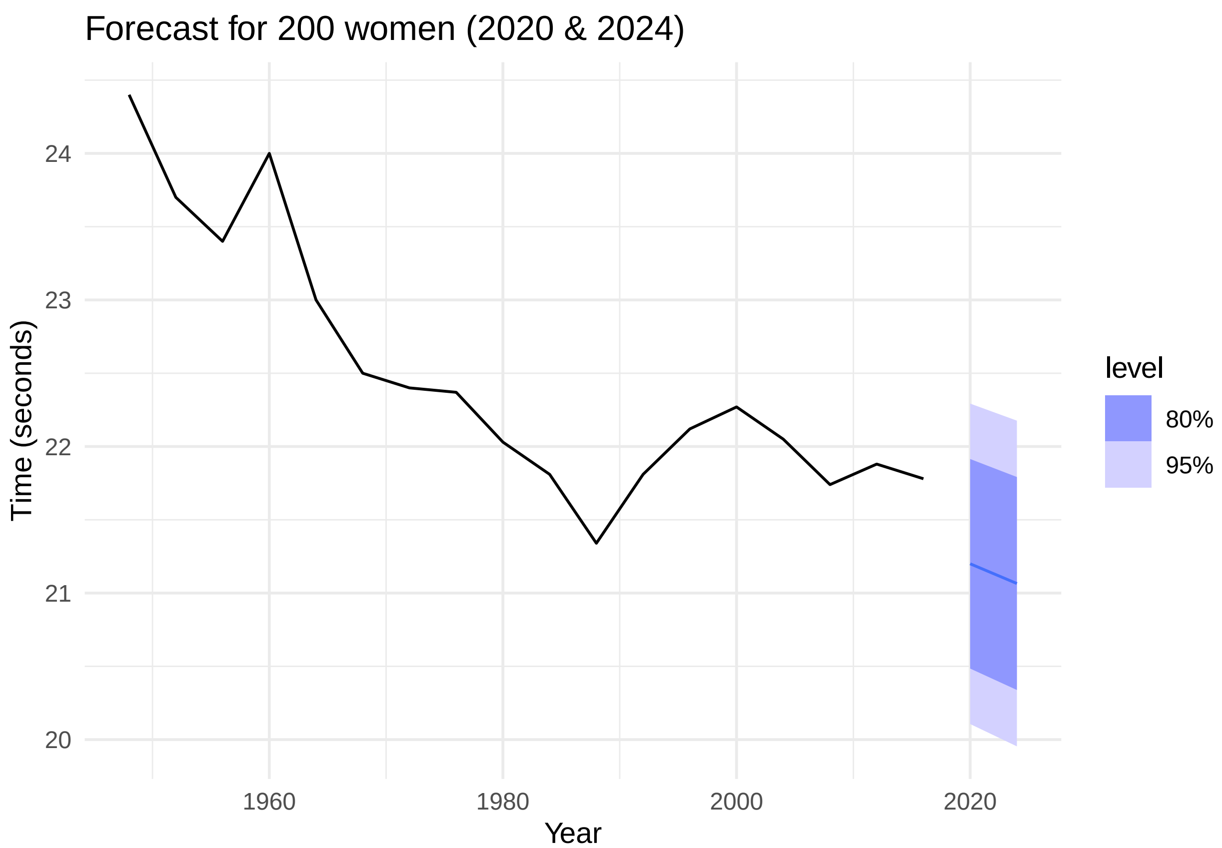


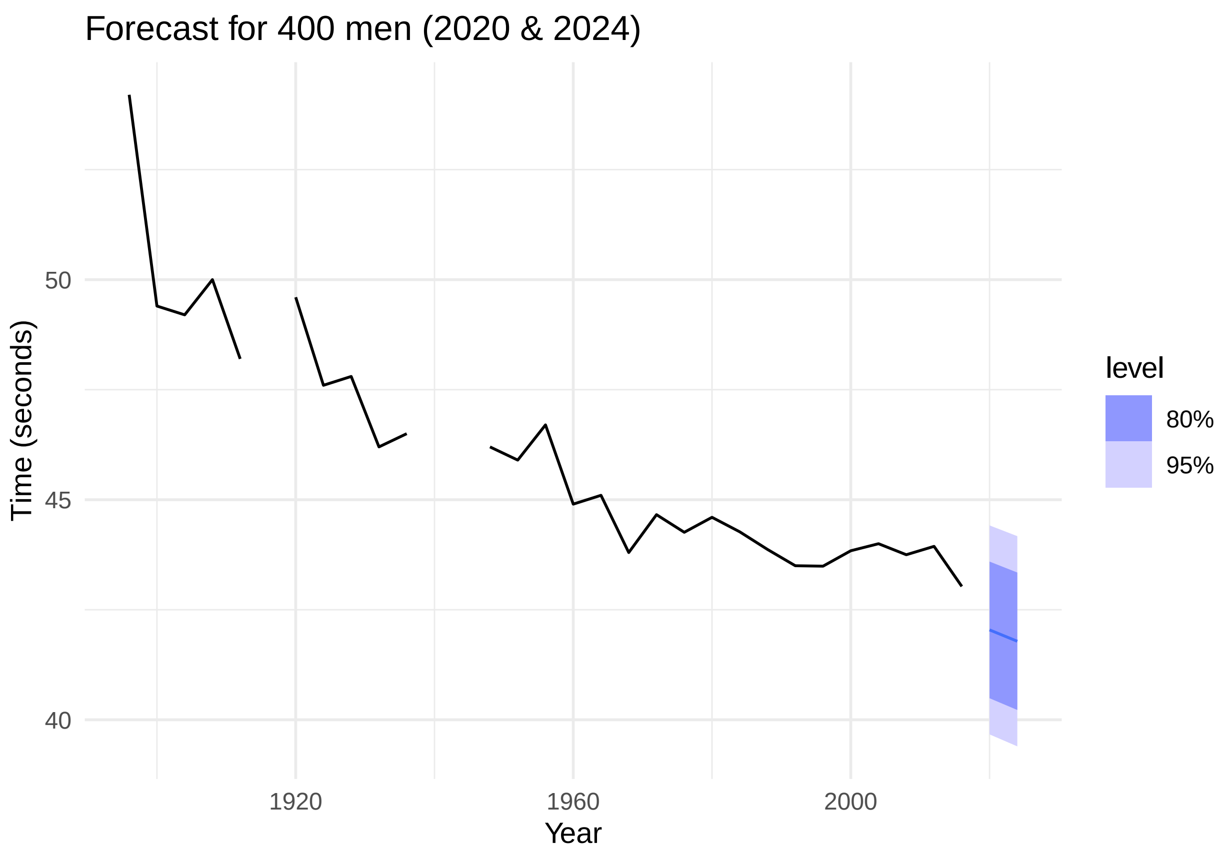
Forecast for 10000 women (2020 & 2024)
Time (seconds)
1850
1800
1750
1700
1990
2000
2010
2020
Year
level
80%
95%

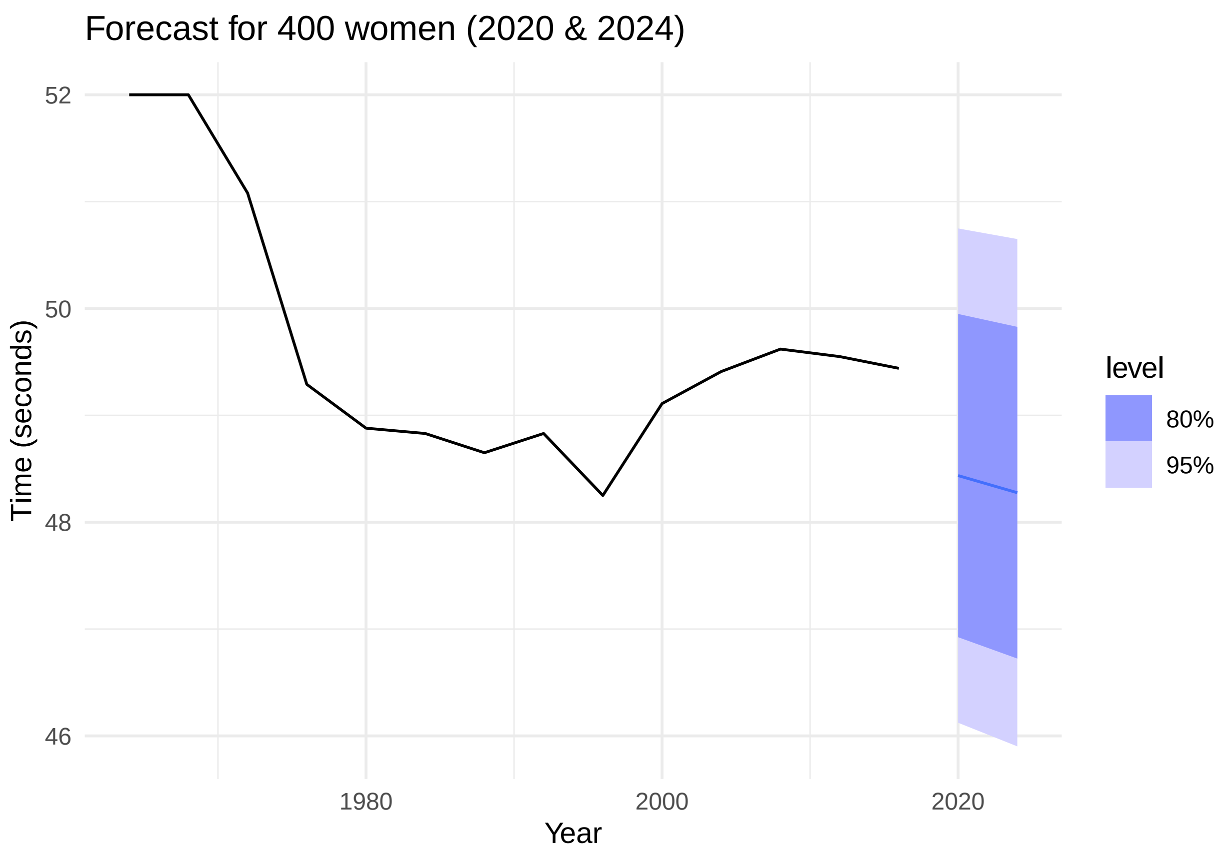

Forecast for 1500 men (2020 & 2024)
Time (seconds)
260
240
220
200
1920
1960
2000
Year
level
80%
95%

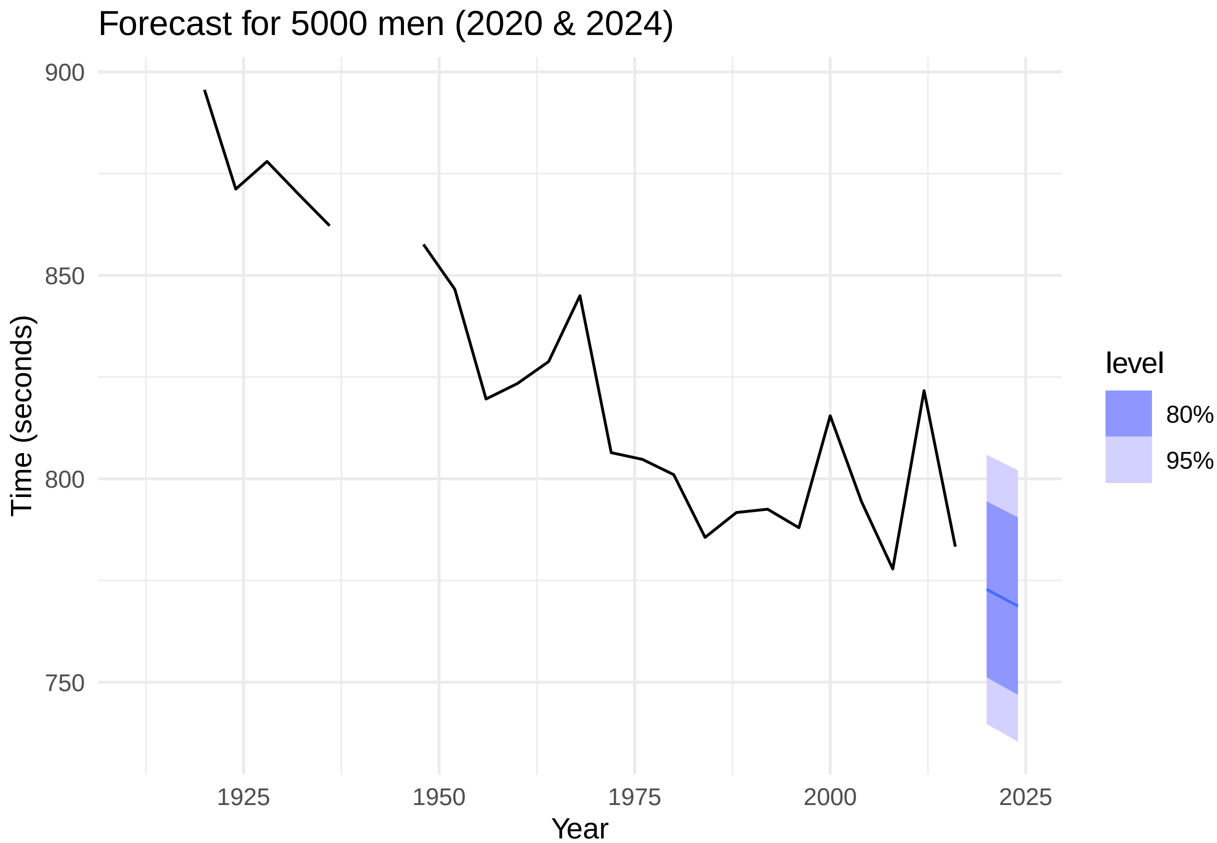

Forecast for 1500 women (2020 & 2024)
Time (seconds)
250
240
1970
1980
1990
2000
2010
2020
Year
level
80%
95%


Forecast for 200 men (2020 & 2024)
Time (seconds)
22
21
20
19
1920
1960
2000
Year
level
80%
95%

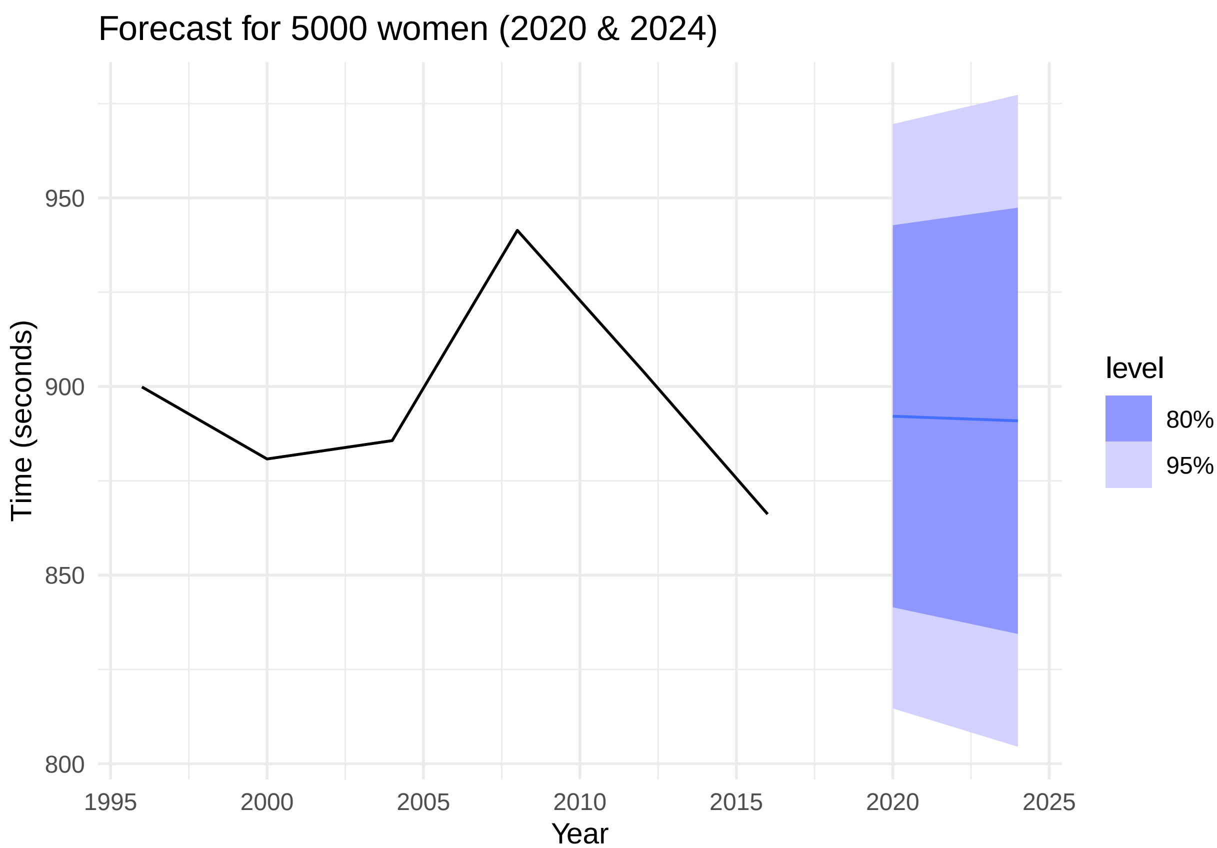

Forecast for 200 women (2020 & 2024)
Time (seconds)
24
23
22
21
20
1960
1980
2000
2020
Year
level
80%
95%

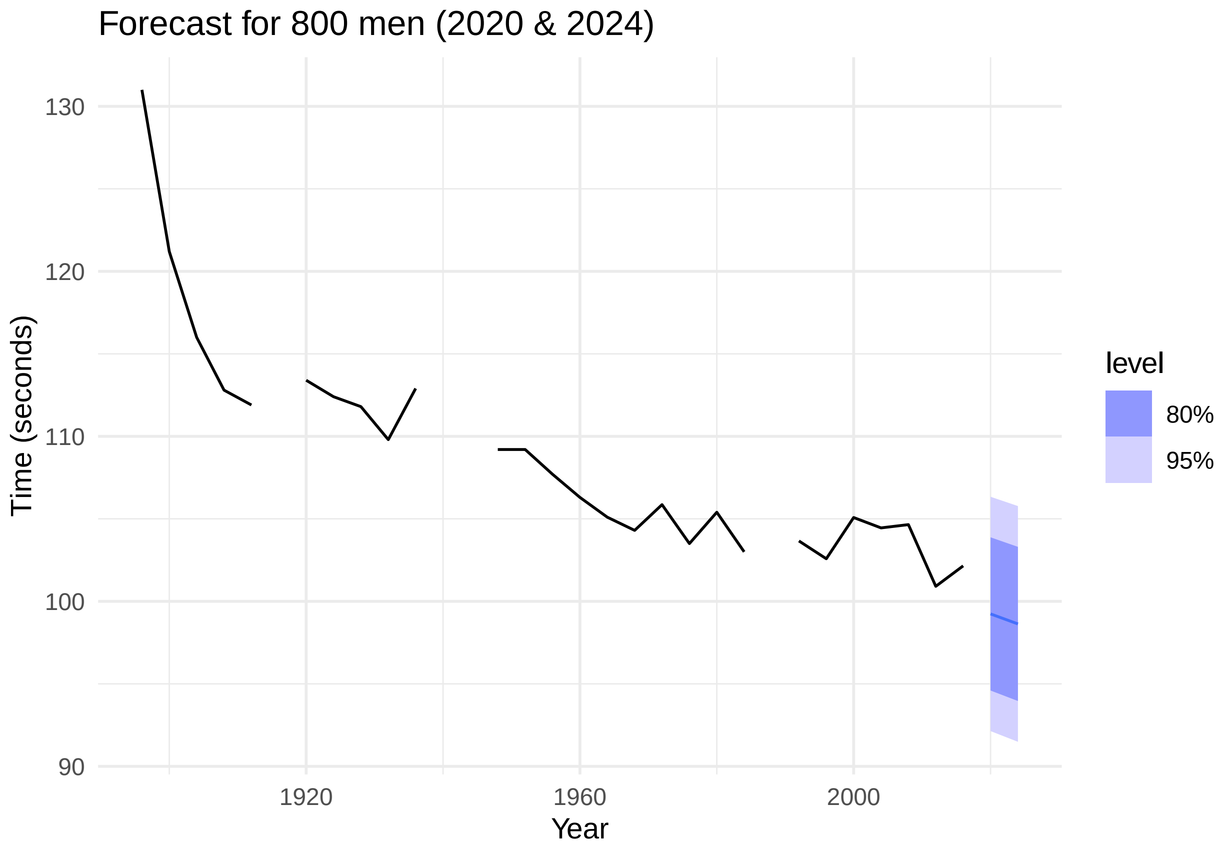

Forecast for 400 men (2020 & 2024)
Time (seconds)
50
45
40
1920
1960
2000
Year
level
80%
95%

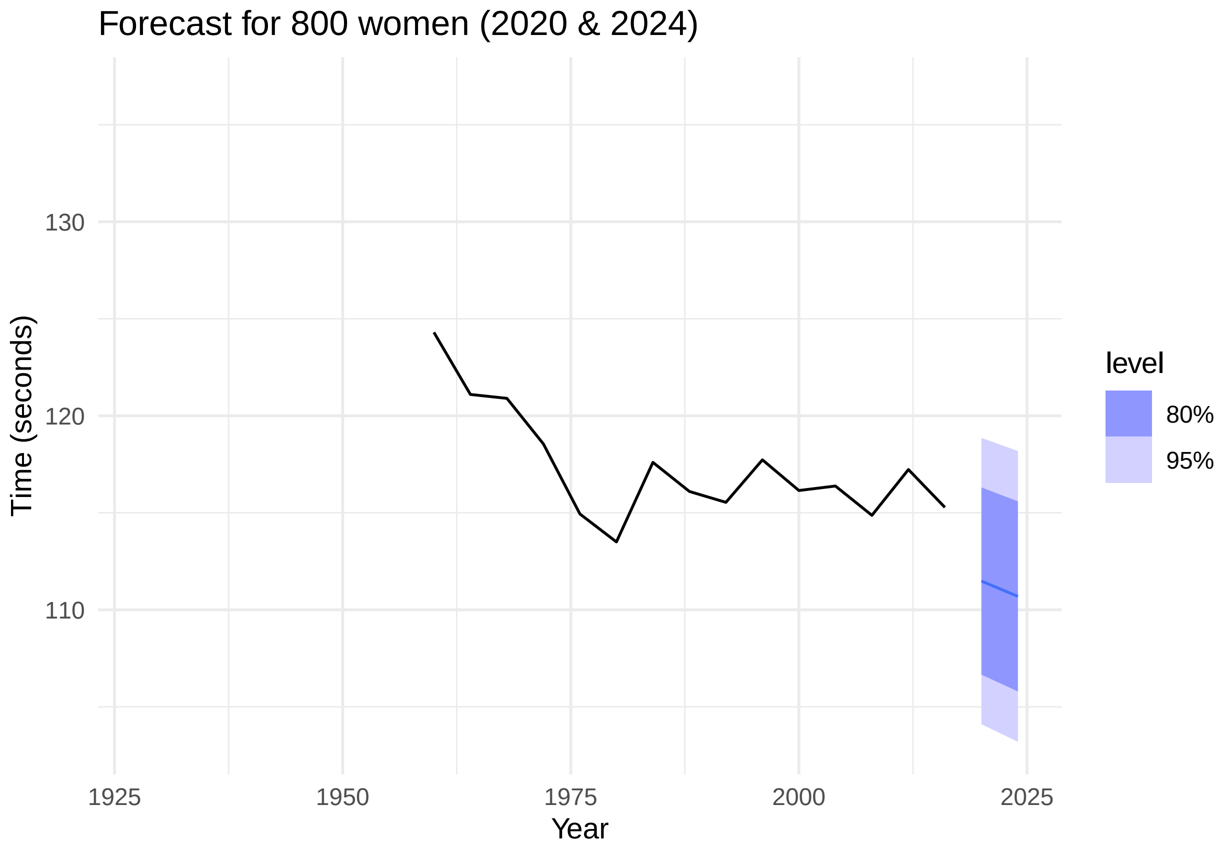

Forecast for 400 women (2020 & 2024)
52
50
48
46
Time (seconds)
1980
2000
2020
Year
level
80%
95%



Forecast for 5000 men (2020 & 2024)
900
850
800
750
Time (seconds)
1925
1950
1975
2000
2025
Year
level
80%
95%


Forecast for 5000 women (2020 & 2024)
Time (seconds)
950
900
850
800
1995
2000
2005
2010
2015
2020
2025
Year
level
80%
95%


Forecast for 800 men (2020 & 2024)
130
120
110
100
90
1920
1960
2000
Year
Time (seconds)
level
80%
95%



Question 2: Elasticity Coefficient in Log-Log Model

An elasticity coefficient is the ratio of the percentage change in the forecast variable y to the percentage change in the predictor variable x. Mathematically, the elasticity is defined as $\frac{dy}{dx} \cdot \frac{x}{y}$.

The question asks us to consider the log-log model: $\log(y_t) = \beta_0 + \beta_1 \log(x_t) + \epsilon_t$ and show that β_1 is the elasticity coefficient.

Solution:

First, I'll express y as a function of x by working with the log-log model equation:

$\log(y_t) = \beta_0 + \beta_1 \log(x_t) + \epsilon_t$

To isolate y_t, I need to exponentiate both sides:

$y_t = \exp(\beta_0 + \beta_1 \log(x_t) + \epsilon_t)$

Using the properties of exponentials:

$y_t = \exp(\beta_0) \times \exp(\beta_1 \log(x_t)) \times \exp(\epsilon_t)$

Since $\exp(\beta_1 \log(x_t)) = x_t^{\beta_1}$, I get:

$y_t = \exp(\beta_0) \times x_t^{\beta_1} \times \exp(\epsilon_t)$

Now, to show that β_1 is the elasticity coefficient, I need to calculate: elasticity $= \frac{dy}{dx} \cdot \frac{x}{y}$

First, I'll find $\frac{dy}{dx}$ by differentiating y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}[\exp(\beta_0) \times x_t^{\beta_1} \times \exp(\epsilon_t)]$

Since $\exp(\beta_0)$ and $\exp(\epsilon_t)$ are constants with respect to x: $\frac{dy}{dx} = \exp(\beta_0) \times \exp(\epsilon_t) \times \frac{d}{dx}[x_t^{\beta_1}]$ $\frac{dy}{dx} = \exp(\beta_0) \times \exp(\epsilon_t) \times \beta_1 \times x_t^{(\beta_1 - 1)}$

Now, I'll calculate the elasticity: elasticity $= \frac{dy}{dx} \cdot \frac{x}{y}$ elasticity $= [\exp(\beta_0) \times \exp(\epsilon_t) \times \beta_1 \times x_t^{(\beta_1 - 1)}] \times [\frac{x_t}{(\exp(\beta_0) \times x_t^{\beta_1} \times \exp(\epsilon_t))}]$

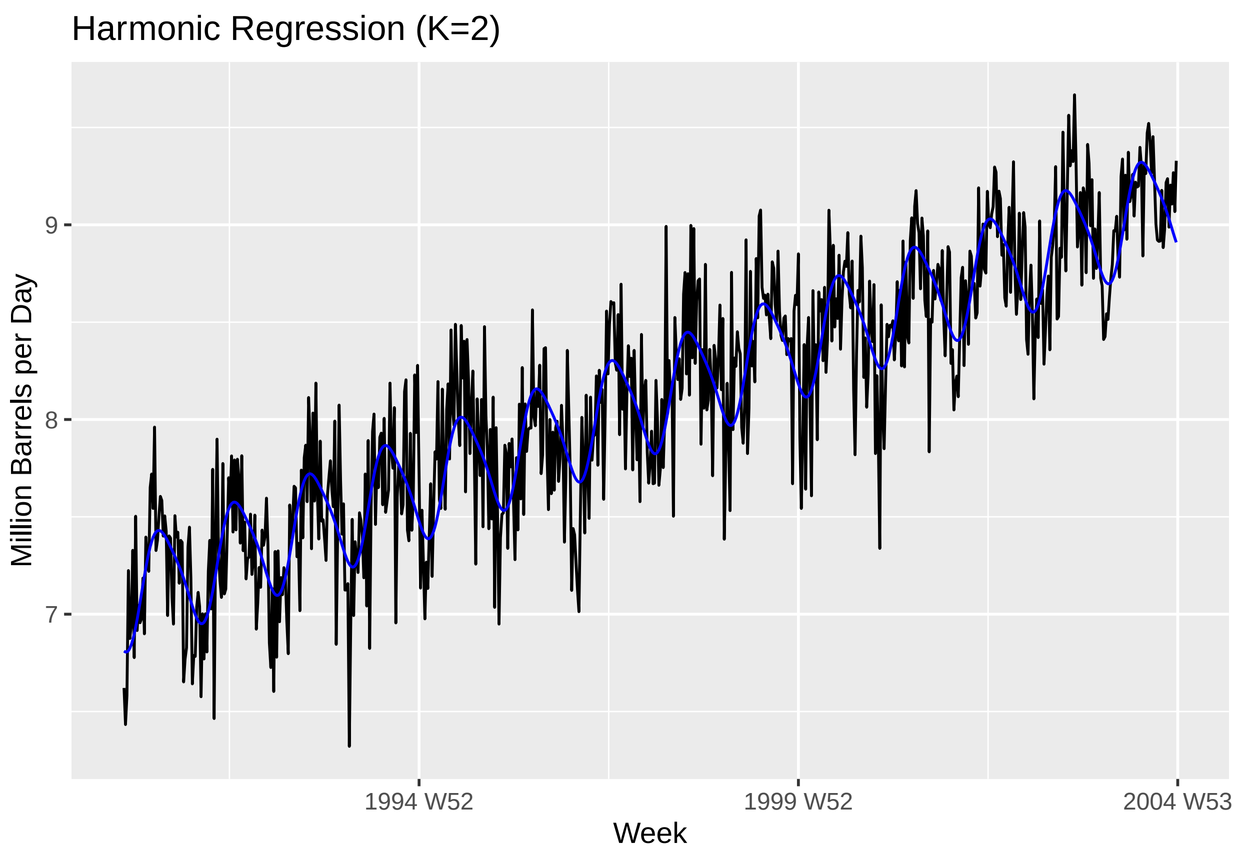
Simplifying: elasticity $= \beta_1 \times x_t^{(\beta_1 - 1)} \times \frac{x_t}{x_t^{\beta_1}}$ elasticity $= \beta_1 \times x_t^{(\beta_1 - 1 + 1 - \beta_1)}$ elasticity $= \beta_1 \times x_t^0$ elasticity $= \beta_1$

Therefore, in a log-log model, the coefficient β_1 represents the elasticity of y with respect to x.

Question 3: US Gasoline Time Series

3.1 Fit harmonic regression with trend

```
data("us_gasoline")

gasoline_2004 <- us_gasoline %>%
  filter(Week <= yearweek("2004 W52"))

harmonic_model_k2 <- gasoline_2004 %>%
  model(TSLM(Barrels ~ trend() + fourier(period = 52, K = 2)))

harmonic_model_k2 %>%
  augment() %>%
  ggplot(aes(x = Week)) +
  geom_line(aes(y = Barrels), color = "black") +
  geom_line(aes(y = .fitted), color = "blue") +
  labs(title = "Harmonic Regression (K=2)", y = "Million Barrels per Day", x = "Week")
```



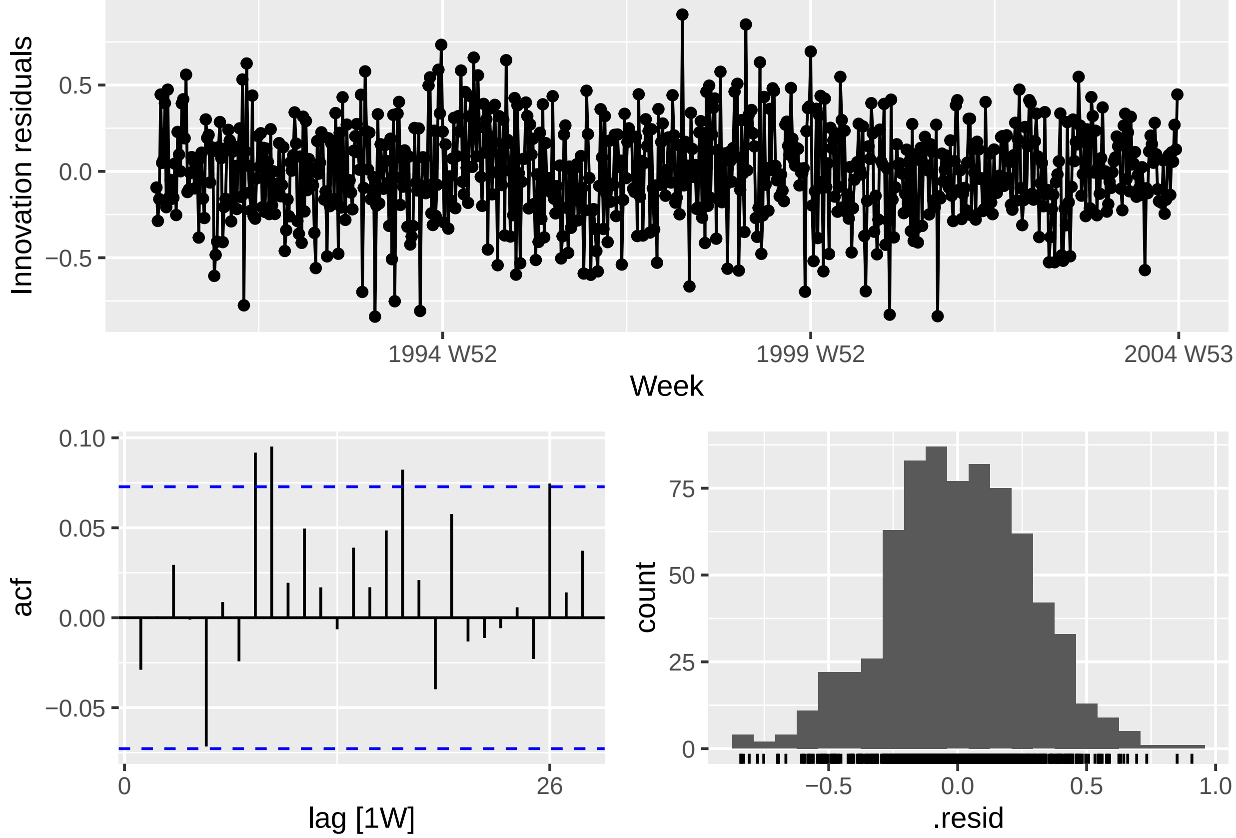
3.2 Select best number of Fourier terms by AICc

```
aicc_values <- tibble(K = 1:6) %>%
  mutate(
    model = map(K, ~ {
      gasoline_2004 %>%
        model(TSLM(Barrels ~ trend() + fourier(period = 52, K = .x)))
    }),
    AICc = map_dbl(model, ~ glance(.x)$AICc)
  )

aicc_values
```

```
## # A tibble: 6 x 3
##       K model              AICc
##   <int> <list>            <dbl>
## 1     1 <mdl_df [1 x 1]> -1808.
## 2     2 <mdl_df [1 x 1]> -1813.
## 3     3 <mdl_df [1 x 1]> -1852.
## 4     4 <mdl_df [1 x 1]> -1856.
## 5     5 <mdl_df [1 x 1]> -1862.
## 6     6 <mdl_df [1 x 1]> -1883.
```

3.3 Residual diagnostics and Ljung-Box test

```r
best_model <- gasoline_2004 %>%
  model(TSLM(Barrels ~ trend() + fourier(period = 52, K = 3)))

best_model %>%
  gg_tsresiduals()
```




```r
augment(best_model) %>%
  features(.resid, feat = ljung_box, lag = 24, dof = 6)
```

```
## # A tibble: 1 x 3
##   .model                                             lb_stat lb_pvalue
##   <chr>                                                <dbl>     <dbl>
## 1 TSLM(Barrels ~ trend() + fourier(period = 52, K = 3))   33.0    0.0165
```

3.4 Forecast for 2005

```r
forecast_2005 <- best_model %>%
  forecast(h = "1 year")

forecast_2005 %>%
```

```r
autoplot(us_gasoline) +
labs(title = "Forecast of U.S. Gasoline Supply for 2005",
     y = "Million Barrels per Day", x = "Year")
```